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Algorithmic probability

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Question: Is the restriction to prefix programs sufficient to define algorithmic probability?

Answer: Yes. Prefix programs are exclusive of each other, so their probabilities sum up to less than 1.

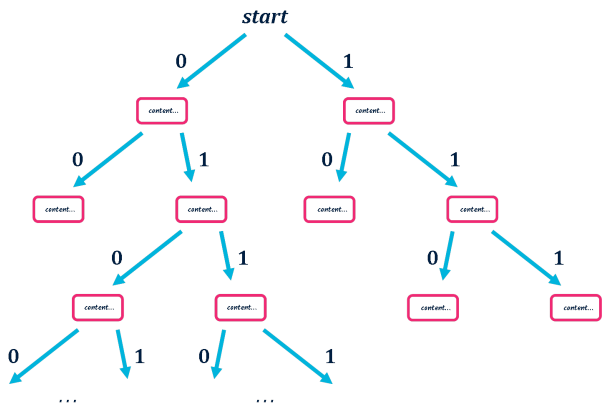
We will see below that **prefix programs** can be represented as leaves in a binary tree. The deeper the leaf, the less probable the program. More importantly, all programs together span at most the whole tree, so their cumulative probability is at most 1. **Prefix complexity** can therefore be used to define algorithmic probability by making the initial suggestion made by Ray Solomonoff in 1964 rigorous. The point is to add up the probabilities $2^{-|p|}$ of all prefix programs p that generate object s .

Algorithmic probability

$$P(s) = \sum_{p: U_{pr}(p)=s} 2^{-|p|}$$

► See why the series in the definition of algorithmic probability always converges

In the next exercise, we consider programs that explore a binary tree. Such programs choose the next branch to the left if their next digit in the input is 0, and the next branch to the right if their next digit is 1. A program's outcome is the concatenation of all visited nodes' content.



Binary Tree

1/1 point (ungraded)

In the binary tree example, let's consider that program bits are randomly drawn. Now let's add up the probabilities of all possible outcomes. What is a condition for this sum to be less than 1?

- ☐ The tree is finite.
- ☐ The average branch length is finite.
- ☒ The program asks for more input bits until it reaches a leaf.
- ☐ The program stops when encountering the same n -bit pattern twice in its input, with n given in advance.



Answer

Correct:

If the program asks for more input bits until it reaches a leaf, we are sure that all events are exclusive of each other and that the cumulated probability is less than 1. This is no longer true if the program can stop anywhere in the tree, even if the tree is finite.

Submit

You have used 1 of 1 attempt

✓ Correct (1/1 point)

Since the shortest program generating s makes the greatest contribution to the sum, we may write:

$$P(s) \approx 2^{-K(s)}$$

Shannon’s definition of information for a repeated event s is $I(s) = \log(1/p(s))$, where $p(s)$ is the probability of occurrence of s . If we consider that the occurrences of s can be modelled as the output of a machine fed with random input, then $P(s) \approx p(s)$. This means that for such a repeated event, $I(s) \approx K(s)$. **Algorithmic information** and **Shannon’s information** make contact here, the former being more general than the latter.

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